

Portfolio: Evan Butler

Vehicle Integration /
Dynamics Engineering



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Multi-disciplinary engineering

Full-stack electromechanical, vehicle dynamics, and controls

Data Acquisition:

4 years of design iteration, owning mechanical, electrical, and software systems, integrating into complex and demanding environments.



Active ARB:

Mechanical design: Leveraging intentional compliance towards vehicle dynamics targets, with support for remote actuation.



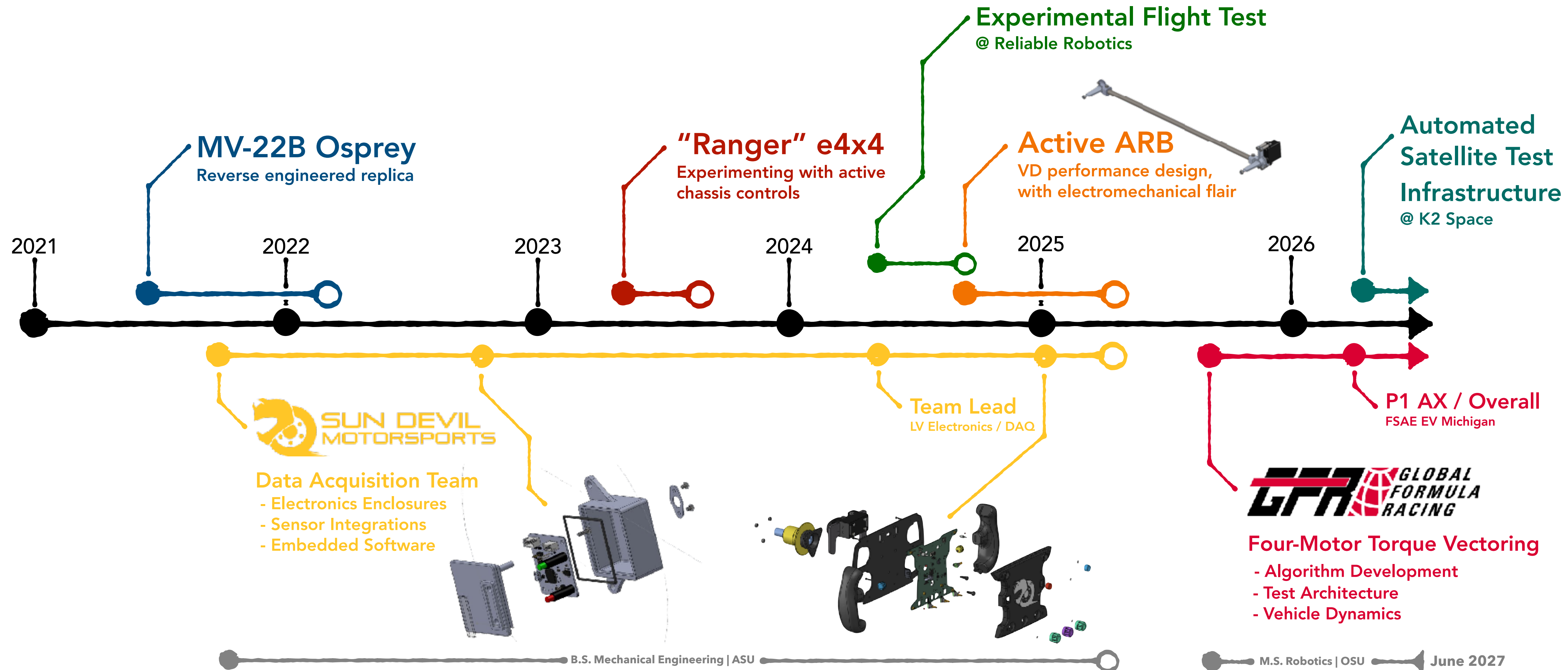
Torque Vectoring:

2 projects in quad-motor control,
first in simulation and literature, then
translating those concepts into real
control architecture.



Timeline

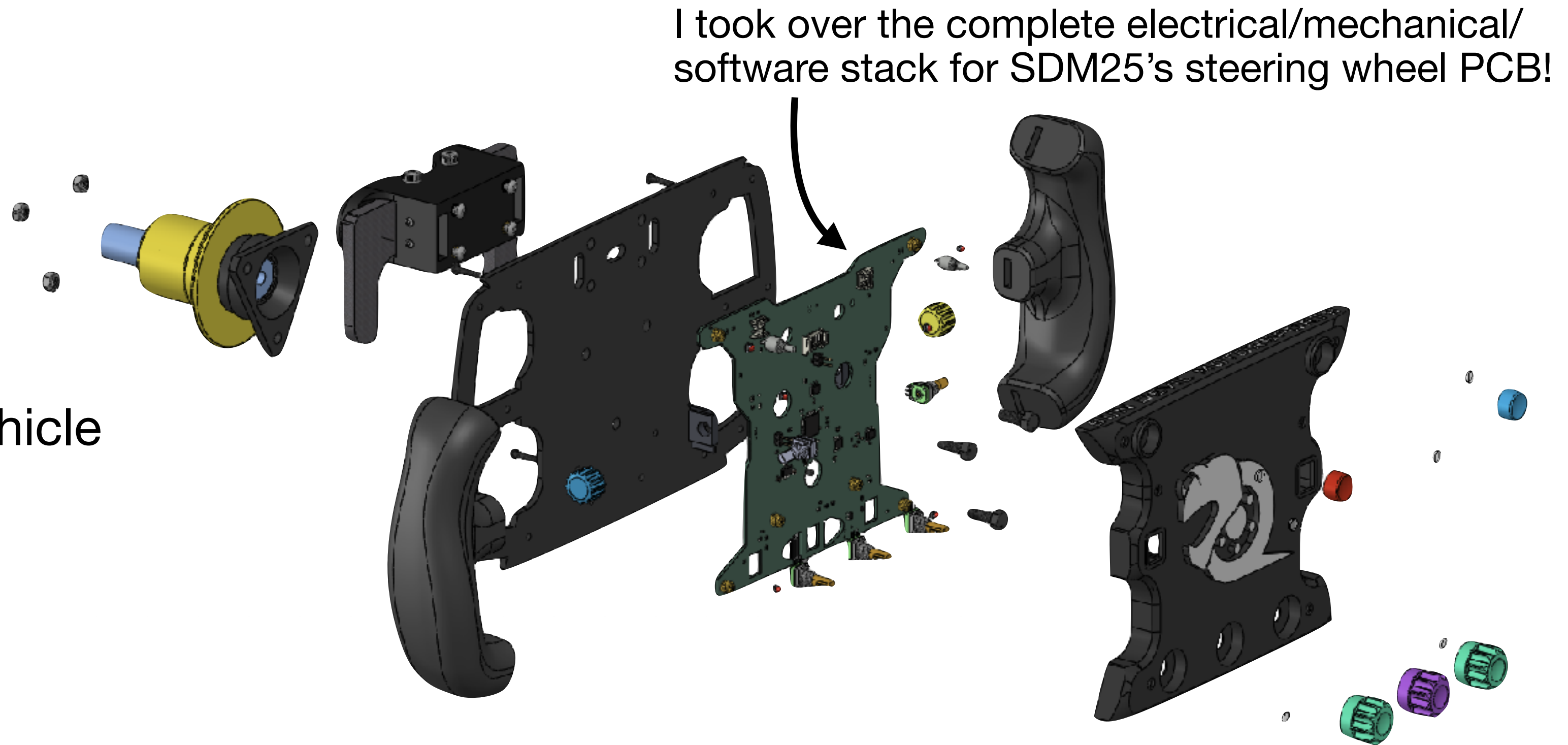
Building, testing, and integrating complex systems.



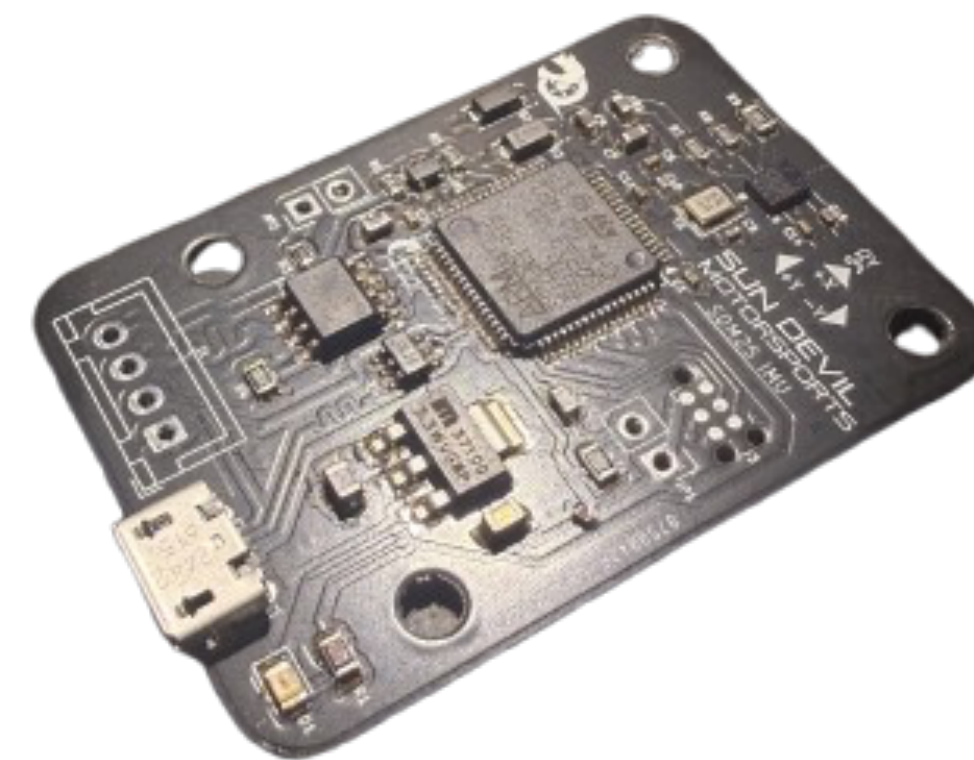
FSAE DAQ

Data acquisition

- Electromechanical sensing, on a vehicle scale, with an eye towards vehicle dynamics;
 - Stakeholder requirements → System requirements.
 - Buildout of a modular, production-optimized architecture.
 - Responsible for LV electronics + enclosures, driver control interfaces, and harnessing.
- Mentored and managed a team of 15 for SDM-25.



I took over the complete electrical/mechanical/
software stack for SDM25's steering wheel PCB!



Converted all custom LV electronics
to a unified STM32 architecture
(IMU pictured)

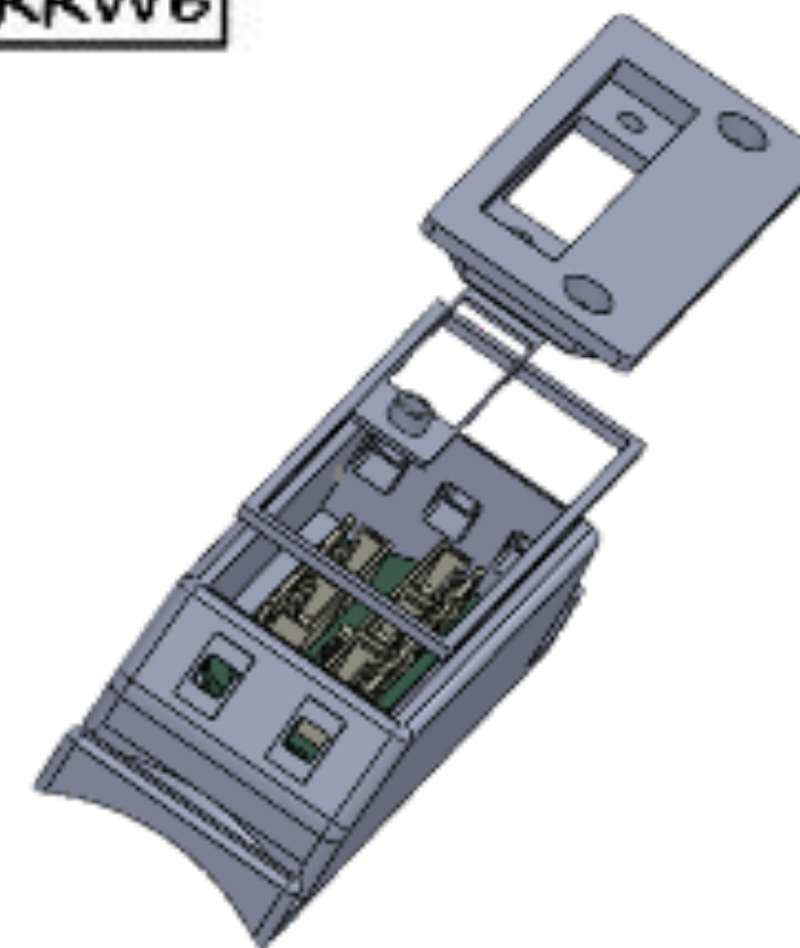
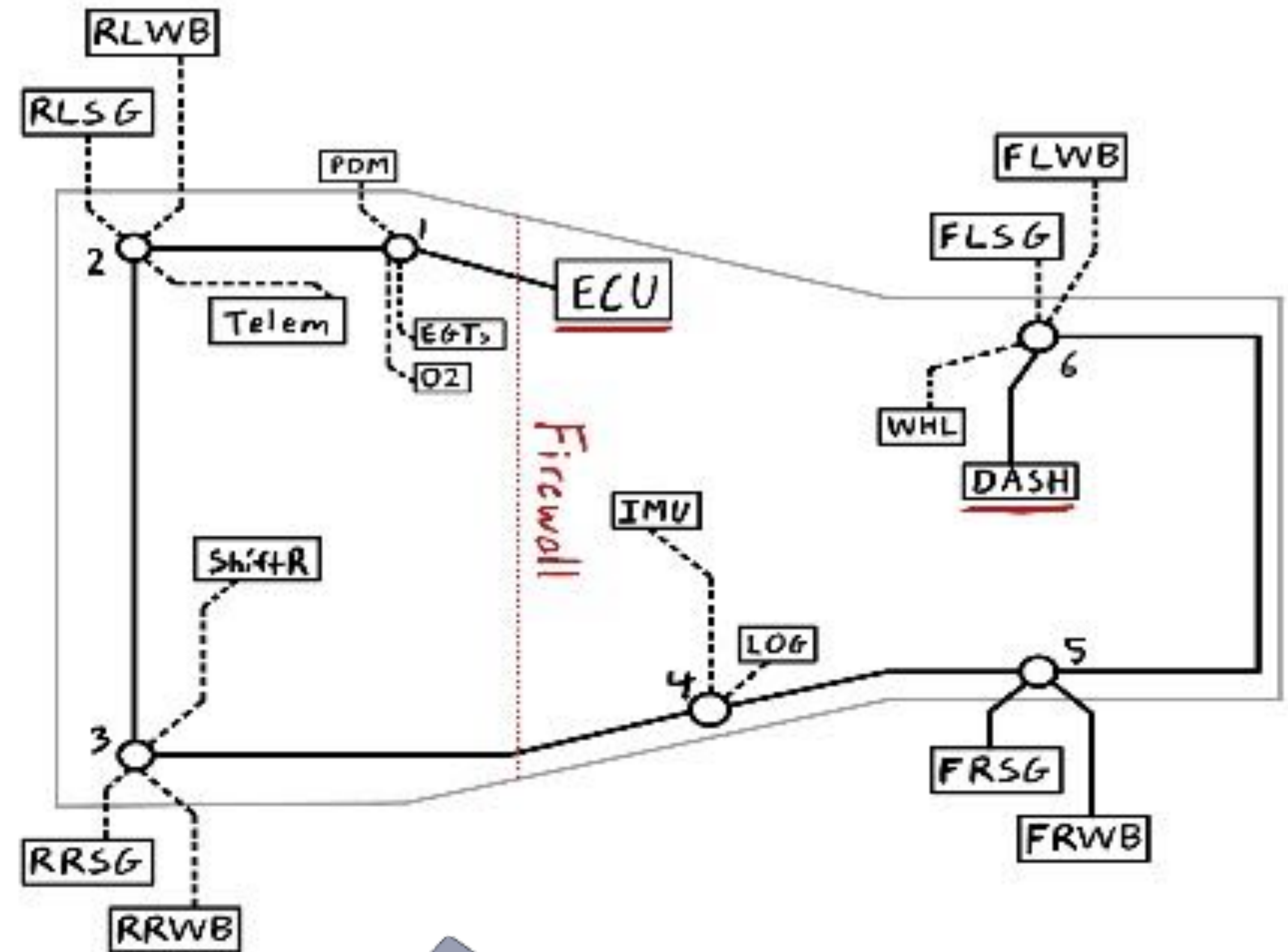


My first part on the team
(shock potentiometer modifications)

FSAE DAQ

System architecture

- Modular node-based CAN, splitters at intervals.
 - Implemented dropout detection, health flags, and diagnostics.
 - Node removal does not have functional penalties.
- Primary system goal was IP55 resistance, achieved through gasketing/sealant on all nodes.
- All data logs to custom datalogger at ~300Hz, sensor variant.
- Optional termination configurations on all nodes.



3rd Generation
CAN Splitter
(SDM25)

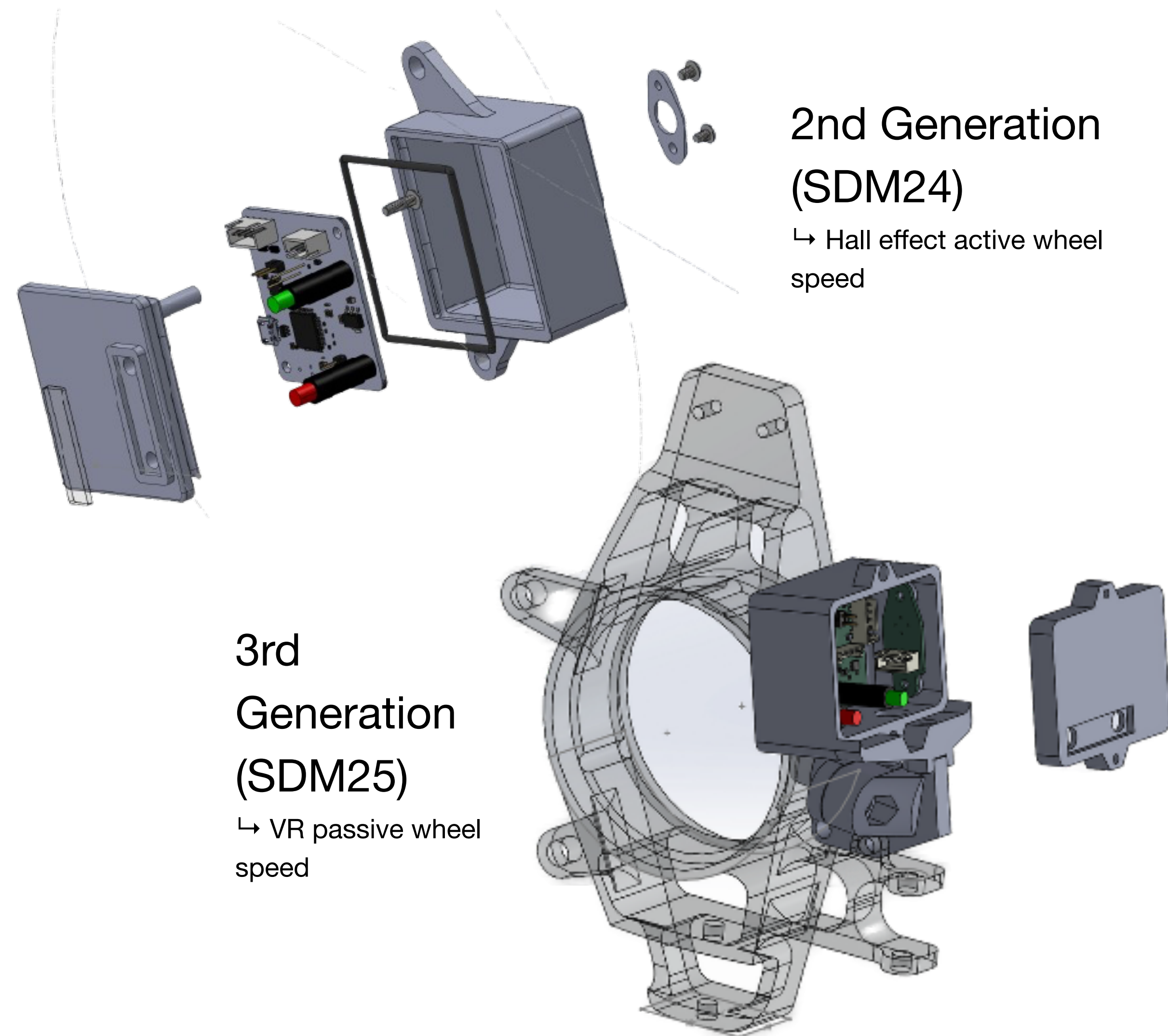
↳ With weather gasketing pictured

FSAE DAQ

Design challenge: In-wheel sensing

- Designed and Integrated measurement systems for **brake rotor temperature** and **wheel speeds**
 - Over 3 generations, localized sensor processing and compute to the outboard.
 - Concept → production with VR-type passive speed sensors.
 - Integrated wheel-speed sensors into bearing spacers, reducing unsprung and rotating inertia, and eliminating rotating parts/fasteners.
 - Validated + tested environmental / structural / thermal design criteria.

System ownership for 3 generations;



Active ARB

Mechanical Design / Vehicle Dynamics

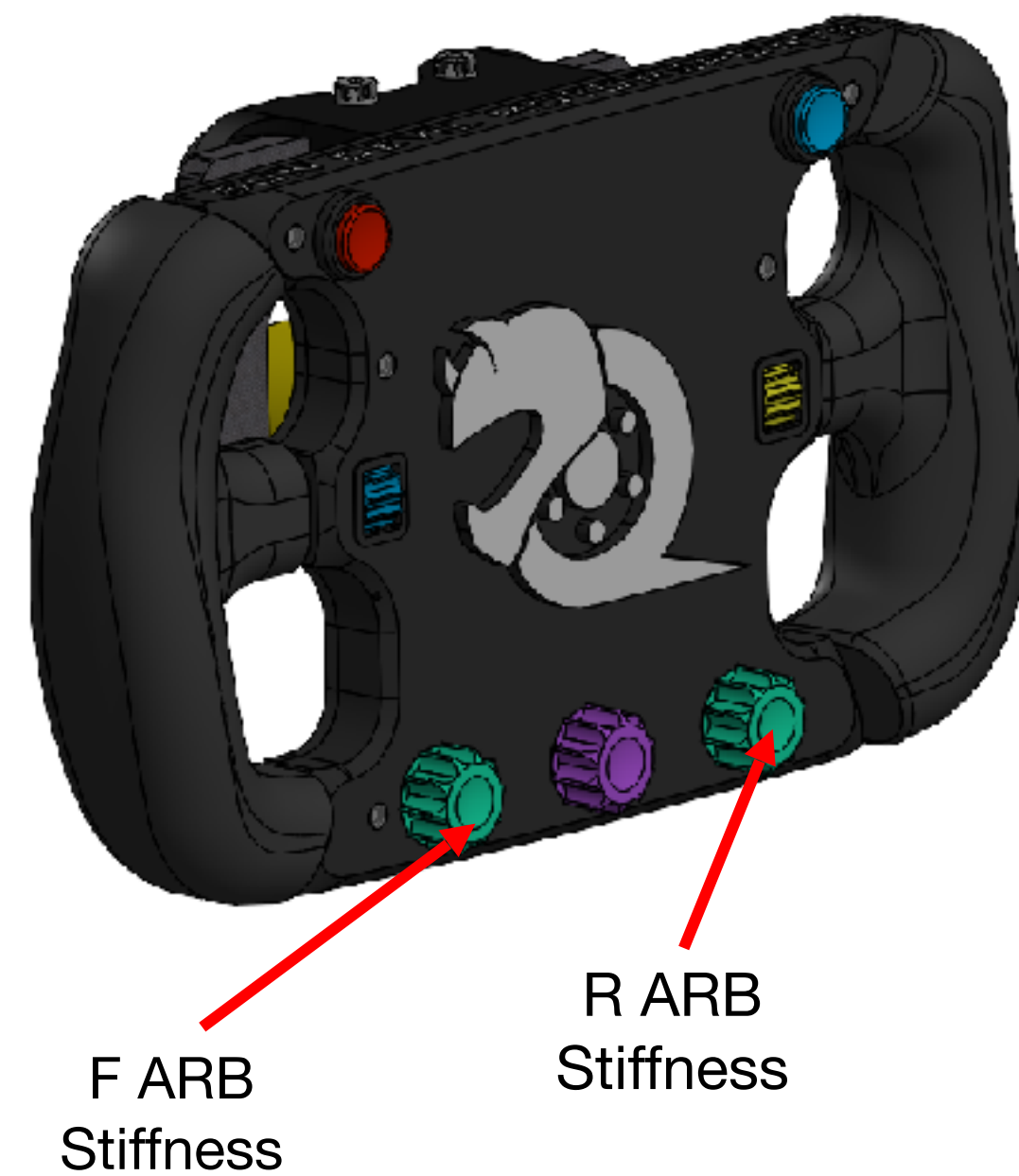
- Blade-style electromechanical adjustable anti-roll bar
 - VD-defined mechanical targets
 - Targeted $.05 \text{ deg} / G$, 4% RSD adjustability
 - Achieved $.03 \text{ deg} / G$, 4% RSD adjustability
 - Excess compliance in pillow blocks led to series spring effect
- Architecture continued on to SDM-26 and SDM-27
- Led/mentored capstone team of 8



Active ARB

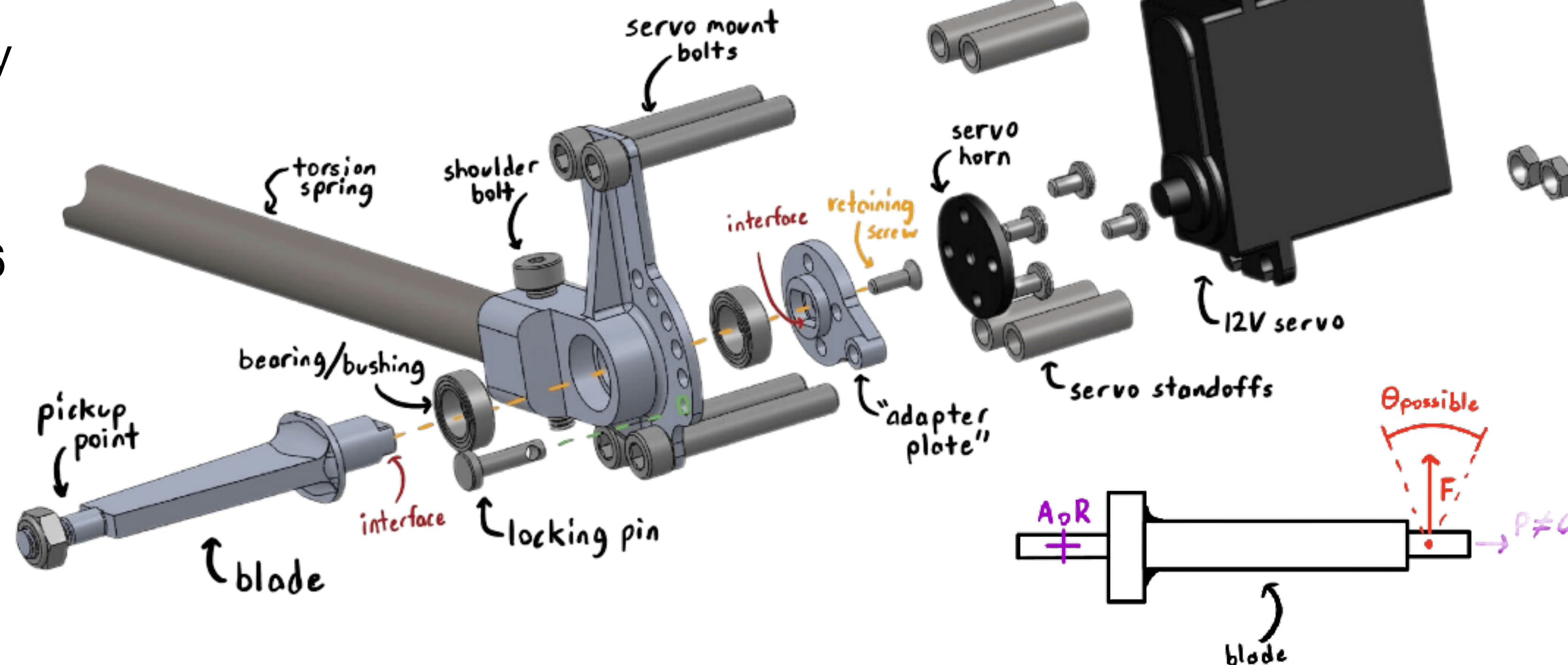
System architecture

- Electronically and manually adjustable configurations
 - 1 servo per ARB, other manually set
- Mixed Steel / AL Componentry
 - Mild steel torsion springs + blocks, 7075 blades
 - Targeted design life of 10^6 cycles in all configurations
- Thorough analyses of induced load cases + operational scenarios



Max Roll Gradient			Min Roll Gradient			Vehicle Inputs		
Client	0.475	deg/G	Roll gradient	0.4	deg/G	Spring Mass	550	lb
Mass Dia	52	%	Roll Stiffness Dia	52	%	CG Height	11.3	in
						RC Height	0.794	in
Roll Stiffness	1377.50	Nm/deg	Total Roll Stiffness	1536.14	Nm/deg			
						Min RSD		
Stiffness	716.45	Nm/deg	F Roll Stiffness	850.76	Nm/deg	F Track	47.5	in
Roll Stiffness	661.34	Nm/deg	Rear Roll Stiffness	785.36	Nm/deg	R Track	47	in
Stiffness	221.45	Nm/deg	F ARB Stiffness	355.82	Nm/deg	Peak A_int	1.25	G
Stiffness	233.31	Nm/deg	R ARB Stiffness	357.31	Nm/deg	F Weight Dist	48	%
W	30.91	%	F ARB %	41.82	%	MR F (Wh/Sk)	1.06	
W	35.28	%	R ARB %	45.50	%	MR R (Wh/Sk)	1.07	
B %	33.01	%	Total ARB %	43.59	%			
						F eLLT	325.46	N
Force	100.01	N	F ARB Force	136.12	N	R eLLT	393.27	N
Force	135.21	N	R ARB Force	174.38	N			
						(linear) F Spring	250	lb/in
						R Spring	225	lb/in
Design Overview						(torsional) F Spring		
Soft Configuration			Stiff Configuration			R Spring		
ss-ARB	171.698	Nm/deg	F Stiffness-ARB	275.774	Nm/deg			
ss-ARB	218.632	Nm/deg	R Stiffness-ARB	329.244	Nm/deg			
Inertia	1313.339	Nm/deg	Total Stiffness	1528.027	Nm/deg			
	0.428	deg/G	Min RSD	46.819	%			
	0.498	deg/G	Max RSD	54.377	%			

System Design / Performance Calculator



Active ARB

Testing + Iterations

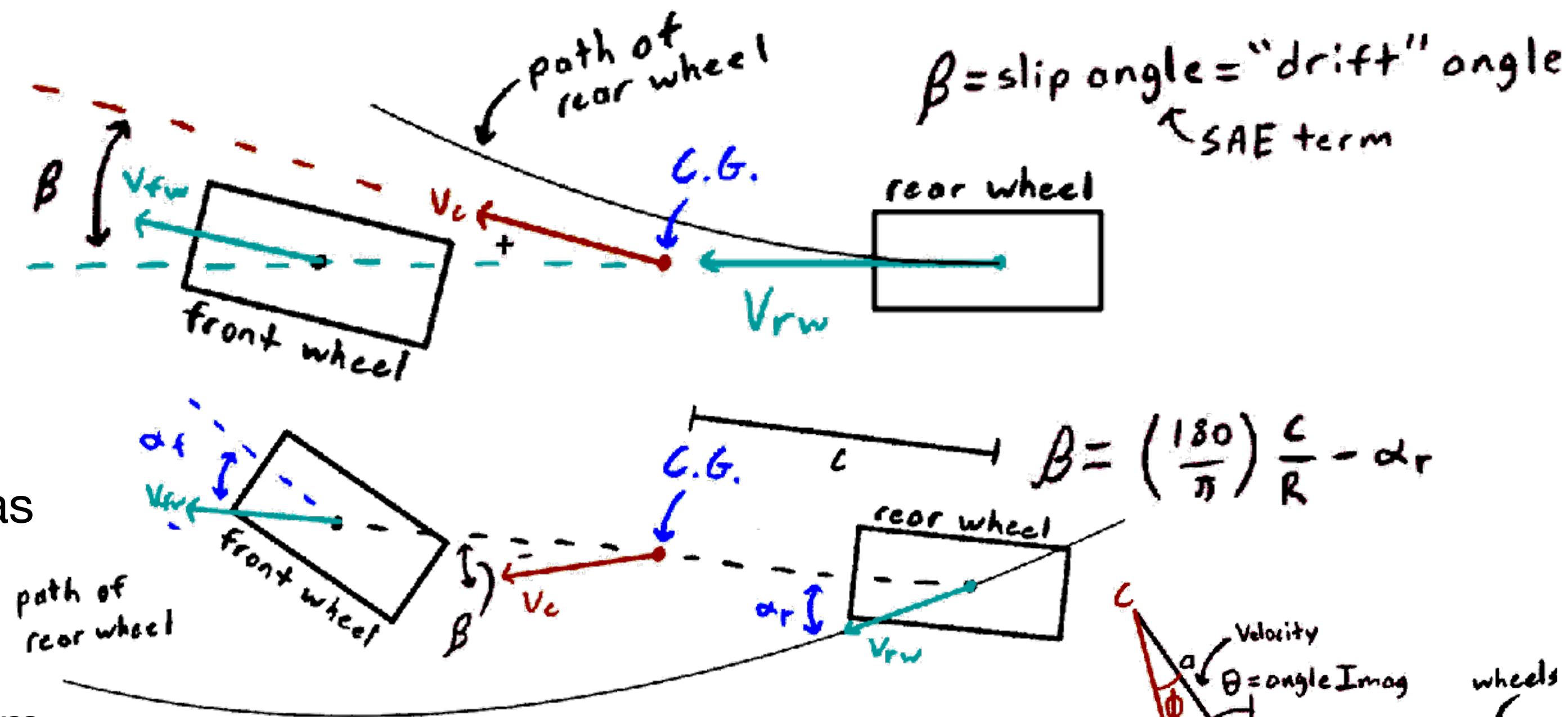
- Conducted component and system level ATPs for validation and traceability.
 - Significant (300%+) excess of compliance found in original ball bearings, replaced with oil-infused bronze bushings to achieve near-target behavior.
 - Servos underwent long-term repeatability tests.
 - Blades and torsion spring passed with minimal error to design and simulated stiffness
- Vehicle-level performance yielded notable performance impacts in skidpad and balance feel.
 - Numerically validated - successful manipulation of RSD and oversteer gradient!



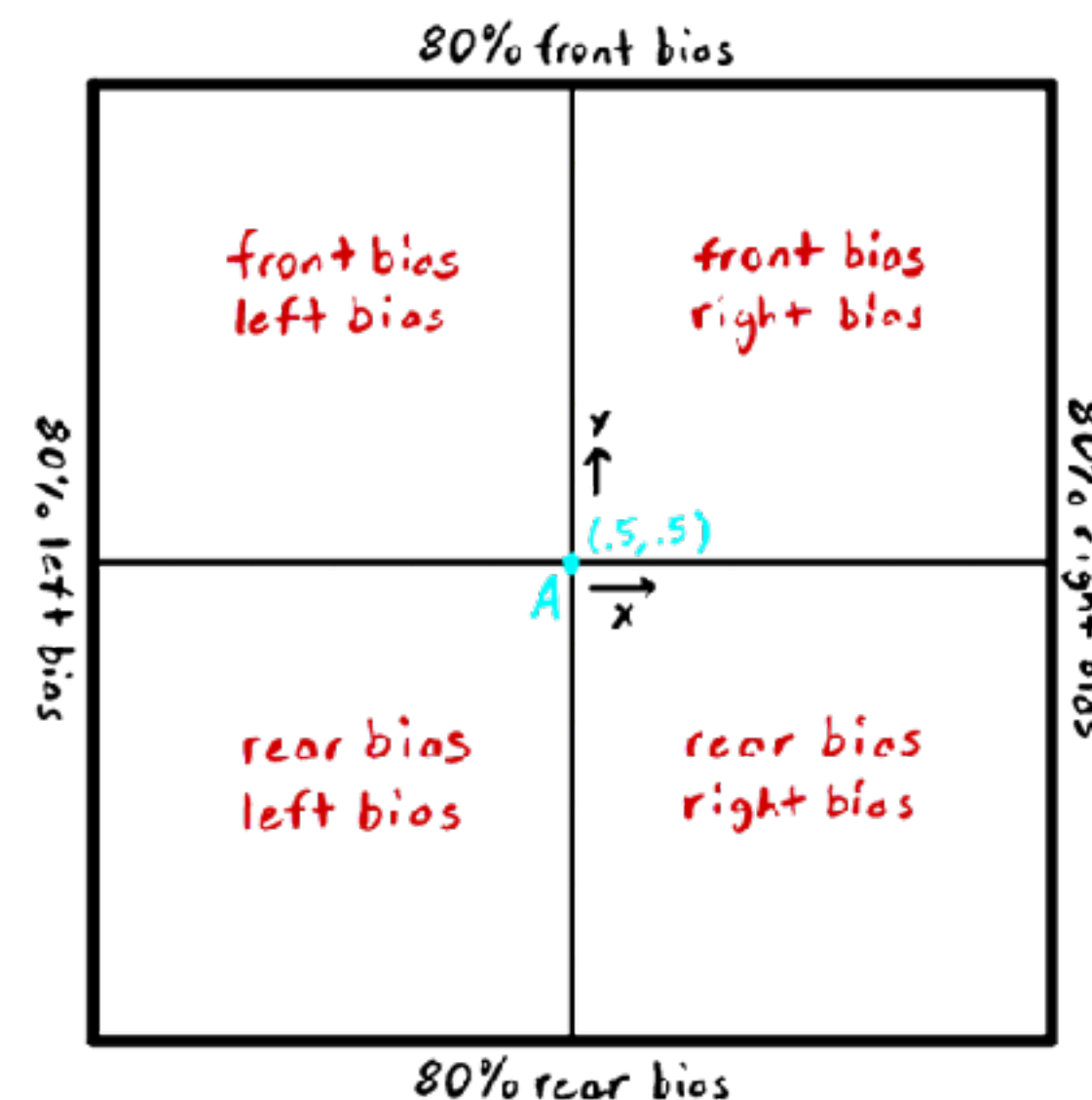
Torque Vectoring

Experimental vehicle dynamics

- Two projects, both centered around vehicle feel as a function of active control.
 - “Ranger” e4x4 - Simulation
 - Designed a real-time traction control system to manage oversteer parameters, utilizing/ self-deriving VD principles from Gillespie.
 - Researched / executed ride analysis/ frequency characterization for my suspension system.
 - GFR FSAE EV - Practical
 - Applied the same concepts to design of a 4-motor TV algorithm for GFR, utilizing my understanding for architecture review and data analysis.



Above: my self-derived calculations before referencing Gillespie



Power Shift Matrix

x axis:

Drift Angle

↳ Function of:

- Steering Angle/Vector
- Velocity Vector

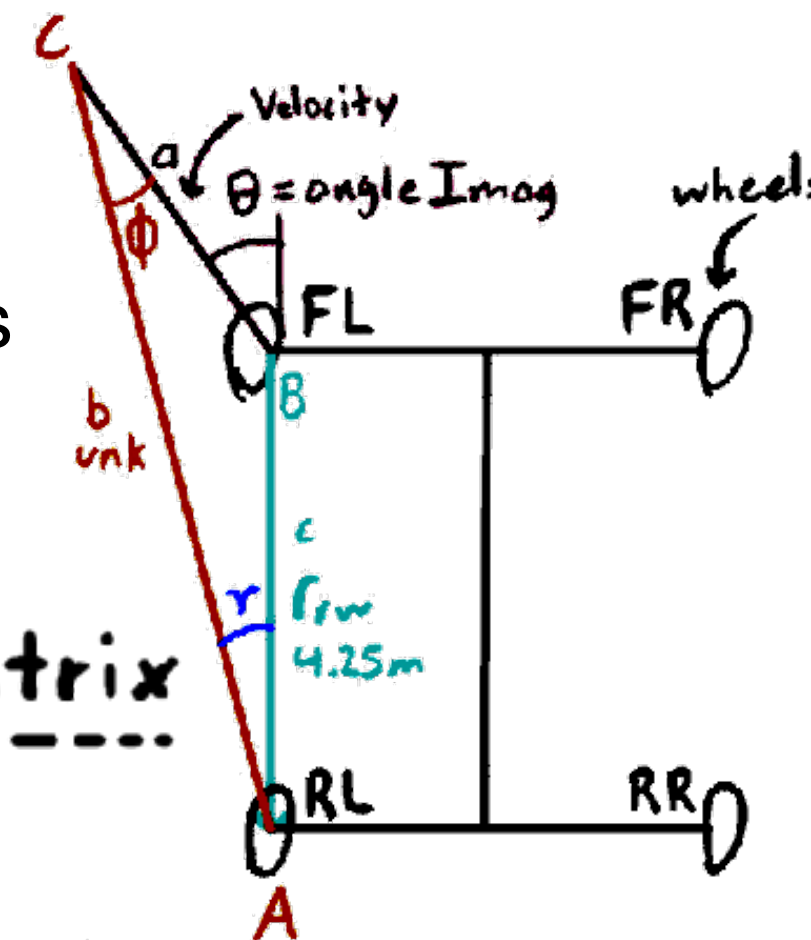
↳ compensation for Δ eLocker

y axis:

Mode Value

Vehicle tilt

static

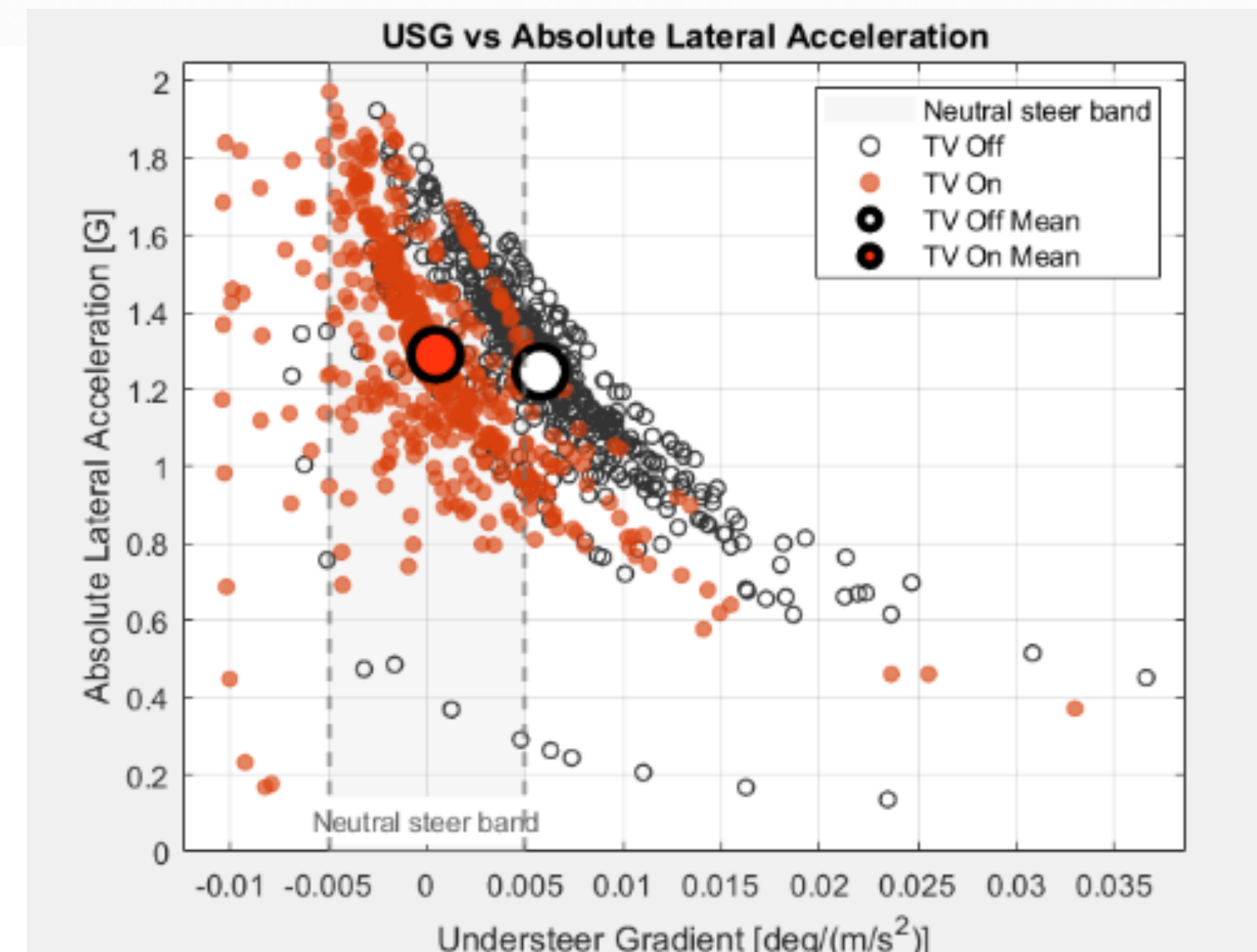
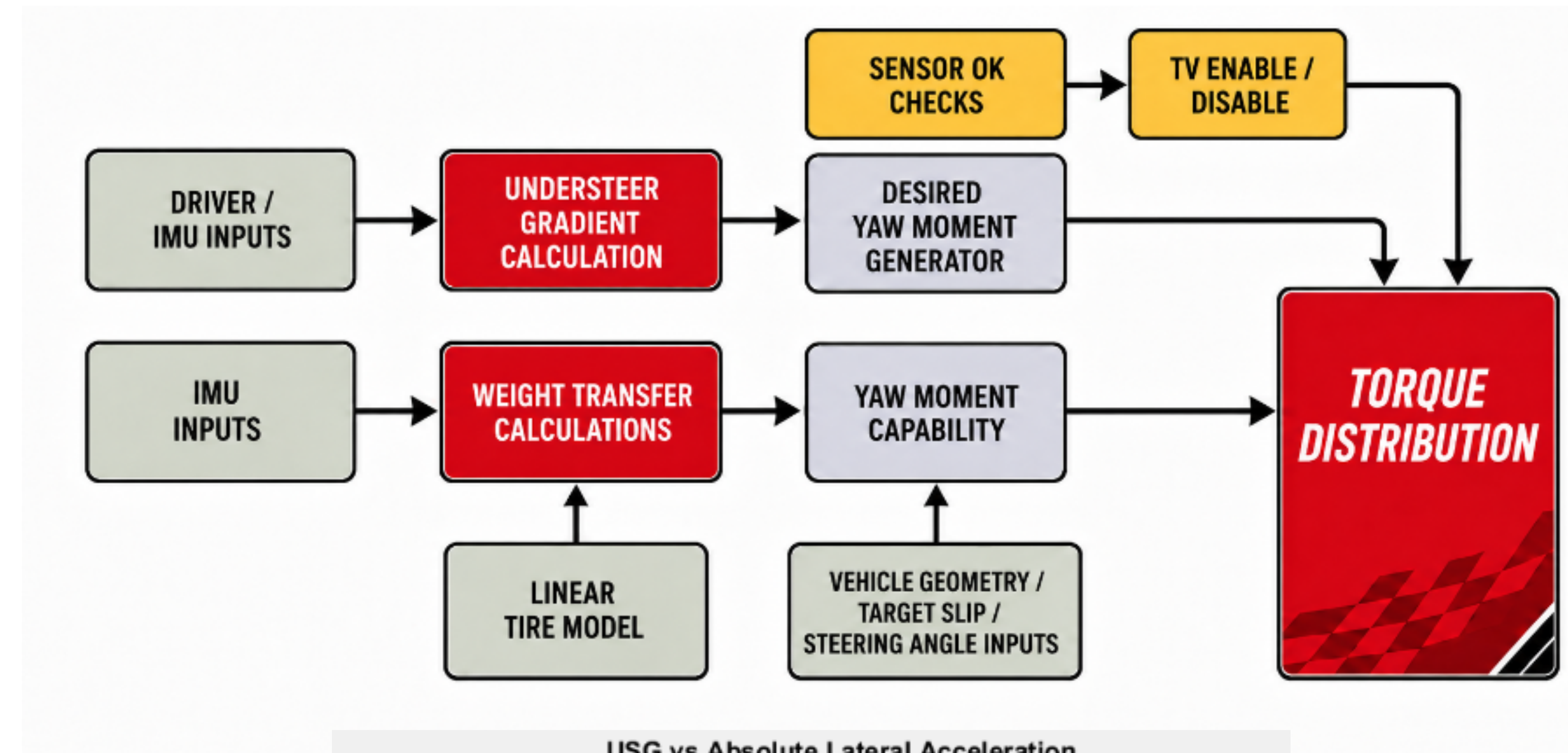


Left: Ranger's power application matrix for torque vectoring

Torque Vectoring

Concepts -> Results

- Concept application with GFR's 4-motor FSAE EV car
 - Steered the architecture design and implementation for our TV model
 - Closed-loop control of instantaneous understeer gradient, with considerations for tire force production capability.
 - Developed test plans and procedures, along with performance expectations and targets.
 - Model yielded a net 3-5% improvement in AX lap times, on average, and P1 in Michigan.



Reliable Robotics

Flight Test / HITL

- Experimental Flight Test
 - Procedure development, execution, and review.
- Hardware-In-The-Loop Support:
 - Sourced and built a functional replica of system-specific hardware for integration into HITL.



I took this photo before a flight test in Mojave, CA!

K2 Space

Ground Test / HITL

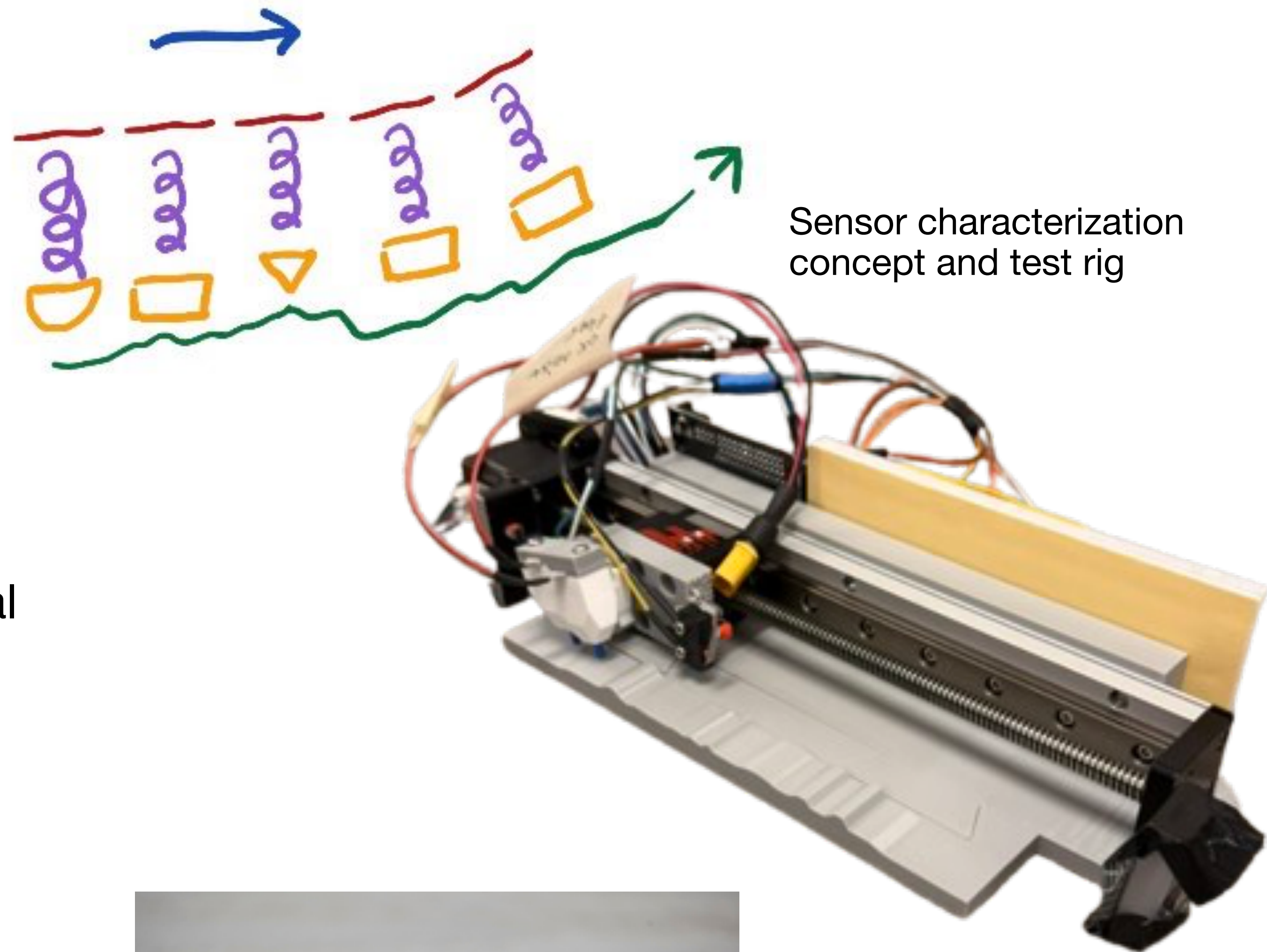
- Ground Test Automation
 - Built out infrastructure and fully automated several engineering tests for technician execution on vehicle.
- Hardware-In-The-Loop:
 - Brought up and validated satellite HITL tables, making them ready for software test.



RAD LAB

Oregon State University

- Studying terrain mechanics, and their interactions with driven and sensing systems.
 - Looking to further human-robot situational awareness through enhanced environmental detection and communication
 - Support ongoing efforts utilizing locomotion (robot steps) to collect rheological (soil performance) measurements utilized to characterize underfoot substrates.



Sensor characterization concept and test rig



LASSIE Project - robotic locomotion rheology

Evan Butler

Integration / Vehicle Dynamics Engineering

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